Week 34: Probability theory

Some practical info:

- Interactive lectures means:
  - I will go over some relevant theory
  - You will have a bit of time to work on exercises
  - I will go over my suggestions for solutions
- Will try to choose problems from previous exams, but you can also suggest problems.
- The timeslot is terrible. Please suggest alternative times that would work for you?
- I am away throughout October, so no lectures then. Will inform on Wiki.

Measurable spaces and probabilities

Informally, a *random experiment* has the following objects:

1. Sample space: Set Ω with possible outcomes $\omega \in \Omega$.
2. Events: Subsets of Ω for which probabilities are defined.
3. Measurable space: Sample space Ω equipped with a family of events $\mathcal{E}$.

Definition 1.1. *(σ-algebra)* A σ-algebra $\mathcal{E}$ on Ω is a family of subsets of Ω satisfying

- $\Omega \in \mathcal{E}$
- If $A \in \mathcal{E}$, then $A^C = \Omega \setminus A \in \mathcal{E}$.
- If $A_1, A_2, \ldots \in \mathcal{E}$, then $\cup_{i=1}^{\infty} A_i \in \mathcal{E}$.

Definition 1.2. *(Events)* The family of events $\mathcal{E}$ is a σ-algebra on Ω.

Definition 1.3. *(Probability)* A probability on $(\Omega, \mathcal{E})$ is a function $P : \mathcal{E} \to \mathbb{R}$ such that

- $P(A) \geq 0$ for all $A \in \mathcal{E}$.
- $P(\Omega) = 1$.
- If $A_1, A_2, \ldots \in \mathcal{E}$ are pairwise disjoint, then $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$.

We often call the tuple $(\Omega, \mathcal{E}, P)$ a *probability space*.

Problem 1

Let $\Omega = \{1, 2, 3\}$ and $\mathcal{E} = \{\emptyset, \Omega, \{1\}, \{2, 3\}\}$.

1. Show that $(\Omega, \mathcal{E})$ is a measurable space.
2. How many subsets of Ω are events? Can you give an example of a set that is not an event?
3. Give an example of a probability P on $(\Omega, \mathcal{E})$.

Random variables

Definition 2.4. *(Random variable)* A random variable (RV) is a function $X : \Omega \to \mathbb{R}$ such that

$$X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{E}$$

for every $B \in \mathcal{B}(\mathbb{R})$. Here $\mathcal{B}(\mathbb{R})$ denotes the Borel σ-algebra on $\mathbb{R}$, which is the smallest σ-algebra containing all open intervals in $\mathbb{R}$.

Problem 2

Let $X : \Omega \to \mathbb{R}$ be such that $(X \leq q)$ is an event for all $q \in \mathbb{Q}$.

1. Prove that $X^{-1}((-\infty, x]) = (X \leq x)$ is an event for every $x \in \mathbb{R}$.
2. Prove that $X^{-1}(I)$ is an event for every interval $I \subseteq \mathbb{R}$.
3. Define $\mathcal{M}_X := \{A \subseteq \mathbb{R} : X^{-1}(A) \in \mathcal{E}\}$, and show that $\mathcal{M}_X$ is a σ-algebra on $\mathbb{R}$.
4. Use the preceding parts to prove that X is a RV.

Problem 3

Let X be a RV. Show that the function $P_X : \mathcal{B}(\mathbb{R}) \to \mathbb{R}$ defined by

$$P_X(B) := P(X \in B) = P\big(X^{-1}(B)\big)$$

is a probability on $\mathbb{R}$.

Expectation

Definition 3.5. *(Simple RV)* A RV X is simple if it takes on only finitely many values. We can write this as $X(\Omega) = \{X(\omega) \mid \omega \in \Omega\} = \{x_1, ..., x_n\}$ for a finite n.

Defining $A_i = \{\omega \in \Omega : X(\omega) = x_i\}$, we have that $A_1, ..., A_n$ is a disjoint partition of Ω, and X can be expressed as $X = \sum_{i=1}^{n} x_i \mathbb{1}_{A_i}$.

Definition 3.6. *(Expectation of simple RVs)* The expectation of a simple RV $X = \sum_{i=1}^{n} a_i \mathbb{1}_{A_i}$ is defined as $\mathbb{E}[X] = \sum_{i=1}^{n} a_i P(A_i)$.

Problem 4

Let X and Y be simple RVs, and let $\alpha, \beta \in \mathbb{R}$. Prove

- Relation to probability: If $A \in \mathcal{E}$, then $\mathbb{E}[\mathbb{1}(A)] = P(A)$.
- Constants are preserved: If $X = c \in \mathbb{R}$, then $\mathbb{E}[X] = c$.
- Monotonicity: If $X \leq Y$, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$.
- Linearity: $\mathbb{E}[\alpha X + \beta Y] = \alpha \mathbb{E}[X] + \beta \mathbb{E}[Y]$.

To conclude…a more general definition of expectation

Thank you for showing up!